Aromal MB

AI/ML Engineer
Chalakudy, Kerala | aromalmb1227@gmail.com | +91-9645405518
linkedin.com/in/aromal-mb | github.com/Aromal-m-b

Summary

AI/ML Engineer who builds classical ML, computer vision and LLM-powered systems end to end, from data preparation and modelling to Docker/FastAPI deployment on AWS. Built 4 agentic and CV projects and placed in 4 hackathons.

Skills

- **Classical ML & Evaluation:** Scikit-learn, Classification, Regression, Clustering, Feature Engineering, Model Evaluation
- **Deep Learning & CV:** PyTorch, TensorFlow, CNN, YOLOv8, CLIP, OpenCV
- **LLMs & Agents:** LangChain, LangGraph, RAG, Prompt Engineering, OpenAI API, Gemini API, Llama 3.3 70B, Ollama, LangSmith
- **Deployment:** Docker, AWS SageMaker, FastAPI, REST APIs, PostgreSQL
- **Data & Tools:** Python, SQL, Pandas, NumPy, Matplotlib, Seaborn, Git/GitHub
- **Soft Skills:** Communication, Problem Solving, Collaboration

Experience

Spectrum Softtech Solutions Pvt. Ltd. – AI/ML Software Trainee *Aug 2026 – Present*

- Tested, evaluated and validated machine learning models across different datasets using hyperparameter tuning.
- Diagnosed edge-case failures and documented variance patterns.

GALTech Technologies Pvt. Ltd. – Intern, Data Science & ML *Jun 2025 – Jul 2025*

- Cleaned and analyzed a dataset of **50,000+** customers and built a clustering model for customer segmentation, boosting targeted campaign response rates by **15%**.

Projects

Pharma App – Agentic Complaint Registration (NLP) *Jun 2026*

- Built an LLM + LangGraph + RAG pipeline (Llama 3.3) that extracts 15 data fields from unstructured text, PDF, DOCX and image complaints.
- Added an interactive chat to edit or complete the form, generates a complaint report on submission, and stores complaints in PostgreSQL via FastAPI.

CityMind – Smart City Management System *Jan – Mar 2026*

- Built a multi-model smart-city platform combining XGBoost, CNN, Gemini and Q-learning under an agentic architecture to make quick city-management decisions; predictions improve as historical data accumulates.

Mugiwara – Learning Disability Detector (NLP) *Jan 2026*

- Built an assessment app for early detection of Dyslexia, Dysgraphia and Dyscalculia in children, with an evaluation report based on each test.

Rasengana – ASL Sign Classification (CNN) *Jan – Feb 2025*

- Trained a PyTorch CNN achieving **90%** accuracy on a 1M+ image dataset; automated dataset labeling and served the model via a FastAPI server.

YOLOv8 Deployment on AWS SageMaker

- Containerized 2 YOLOv8 models with Docker and deployed them on AWS SageMaker (CPU instance, ∼5–8 s inference); monitored cost and latency.

Education

B.Tech in Computer Science Engineering (2022 – 2026), College of Engineering, Poonjar – KTU University CGPA: *7.88*

Achievements

- **Top 5 Finalist** – IHRD AI Samasya, ICGAIFE 3.0
- **First Runner-up** – Gen AI Hackathon (Intra-College), PromptCraft
- **First Runner-up** – IHRD Tharang 2K25 Project Expo
- **Second Runner-up** – Hackengers 2024